

Continuous Curing Oven (IND-OVEN) Equipment Manual and SOP

STANDARD OPERATING PROCEDURE & MAINTENANCE

MANUAL ASSET CLASS: Continuous Curing Oven (IND-OVEN)

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Vindhya Precision Works Engineering Division EFFECTIVE

**DATE: 14 January 2026 APPLICABILITY: Mandatory for all
certified mechanical and electrical maintenance technicians**

1. SECTION 1: OPERATING PARAMETERS AND ALARM THRESHOLDS

1.1 The IND-OVEN asset class shall be operated strictly within the designated parameter bounds defined in Table 1.0. Exceeding these parameters outside the prescribed tolerance windows shall initiate automatic interlock triggers as specified.

Table 1.0: Operating Parameters & Alarm Limits

Parameter	Normal Operating Range	Warning Alarm Setpoint	Critical Interlock / Trip Setpoint	Action on Trip
Zone 1–4 Temperature	185.0°C – 215.0°C	$\pm 3.5^{\circ}\text{C}$ from setpoint	$\pm 7.0^{\circ}\text{C}$ from setpoint	Cut main heating contactor; run exhaust fans at 100% for 30 min.
Conveyor Drive Vibration (RMS)	0.4 mm/s – 1.8 mm/s	2.5 mm/s	4.2 mm/s	Immediate halt of conveyor variable frequency drive (VFD).
Exhaust Duct Static Pressure	-120 Pa to -180 Pa	-95 Pa	-60 Pa	Prevent burner ignition sequence / lock fuel gas solenoid valves.
Recirculation Fan Motor Current	24.5 A – 31.0 A	34.5 A	39.0 A	Tripping of thermal overload relay; alarm to SCADA.

2. SECTION 2: PREVENTIVE MAINTENANCE SCHEDULE

2.1 Preventive maintenance (PM) for IND-OVEN units shall be executed strictly by accumulated running hours recorded via the PLC-104 runtime register.

* Level 1 PM (Every 250 Running Hours $\pm$ 10% Hours):

* Inspect silicone door perimeter seals (Part VPW-SL-099) for splitting or compression set. Replace if clearance gap exceeds 2.0 mm .

* Grease conveyor roller bearings (Pillow Block Type VPW-BR-402) using high-temperature synthetic grease (VPW-GR-888); apply precisely 4 strokes per fitting using a manual grease gun.

* Clean optical safety curtain lenses (Model VPW-OP-12) with isopropyl alcohol wipe.

* Level 2 PM (Every 1,000 Running Hours $\pm$ 25% Hours):

* Verify calibration of all four Type-K thermocouples in Zones 1 through 4 using a certified dry-block calibrator. Replace any sensor showing deviation $> 1.5^{\circ}\text{C}$.

* Inspect internal circulating fan impeller blades (VPW-BL-771) for particulate build-up and dynamic balance. Remove carbon deposits using a non-metallic scraper.

* Check electrical terminal torque inside the main power distribution panel (Cabinet VPW-PNL-02). Retorque all busbar and contactor connections to specifications in Section 5.

* Level 3 PM (Every 5,000 Running Hours $\pm$ 50% Hours):

* Overhaul main exhaust gas damper actuator linkage (VPW-ACT-303). Inspect pivot pins for galling and replace nylon bushings.

* Perform insulation resistance (Megger) test on recirculation fan drive motors (VPW-MOT-55). Minimum acceptable insulation resistance is $100 \text{ M}\Omega$ at 500 V DC .

* Replace internal combustion chamber igniter electrodes and flame scanners.

3. SECTION 3: FAULT-CODE TROUBLESHOOTING TABLE

3.1 When an IND-OVEN fault code appears on the HMI terminal, technicians shall reference Table 2.0 to determine the probable cause and execute the required corrective action.

Table 2.0: Fault-Code Resolution Matrix

Fault Code & Description	Probable Cause	Corrective Action
E-101: Zone Thermal Runaway	Solid-state relay (SSR) failure fused in the closed state, supplying continuous power to heating elements.	Isolate power at main breaker. Test SSR leakage current; replace failed SSR (Part VPW-SSR-21). Verify PID controller output.
E-104: Conveyor Stalls / Over-torque	Mechanical jamming of the wire-mesh belt due to product edge-curl or track misalignment.	Lockout drive (LOTO). Clear jammed product manually. Inspect track guide rollers; adjust tensioner bolts to achieve 15 mm sag at mid-span.
E-202: Low Exhaust Draft Pressure	Accumulation of volatiles and tar in the exhaust ductwork or failure of the exhaust blower motor.	Inspect exhaust stack internal diameter. Scrape out condensates. Check blower belt tension; if slipping, replace belt (Part VPW-BLT-505).
E-305: Burner Flame Failure	Partial blockage of the natural gas pilot orifice or degradation of the UV flame scanner tube.	Clean gas pilot jet with brass wire broach. Clean glass envelope of UV scanner. If fault persists, replace scanner assembly (Part VPW-FLM-88).
E-410: Recirculation Fan Overload	Bearing seizure on the internal circulation fan shaft or choked inlet louvers.	Manually rotate fan shaft (with power locked out). If shaft is bound, replace pillow block bearings (Part VPW-BR-402) and check motor winding balance.
E-520: Safety Interlock Chain Open	Emergency stop button depressed, access door safety switch tripped, or air pressure switch failing.	Inspect safety circuit diagnostic screen on HMI to isolate open contact. Reset or replace faulty microswitch (Part VPW-SW-11); verify door alignment.

4. SECTION 4: CONSUMABLE AND SPARE PARTS LIST

4.1 Technicians shall only use genuine Vindhya Precision Works-approved part numbers listed in Table 3.0 when performing corrective or preventive maintenance on IND-OVEN systems.

Table 3.0: Approved Spares & Consumables

Description	OEM Part Number	Replacement Interval / Stock Unit
High-Temp Silicone Door Seal	VPW-SL-099	Every 1,000 hours / Per Meter
Type-K Thermocouple Assembly	VPW-TC-204	Every 5,000 hours / Unit
Solid-State Relay (50A, 480VAC)	VPW-SSR-21	As needed / Unit
Conveyor Drive V-Belt Set	VPW-BLT-505	Every 2,000 hours / Set of 2
Recirculation Fan Pillow Block Bearing	VPW-BR-402	Every 5,000 hours / Unit
Natural Gas Pilot Igniter Assembly	VPW-IGN-90	Every 2,500 hours / Unit
UV Flame Scanner Tube	VPW-FLM-88	Every 3,000 hours / Unit
High-Temperature Synthetic Grease	VPW-GR-888	Continuous consumable / 400g Cartridge
Exhaust Blower Motor (3.7 kW)	VPW-MOT-55	As needed / Unit
Air Intake HEPA Filter Panel	VPW-FLT-102	Every 500 hours / Unit

5. SECTION 5: TORQUE SPECIFICATIONS

5.1 All threaded fasteners secured during maintenance procedures on the IND-OVEN shall be tightened using a calibrated torque wrench in accordance with Table 4.0.

Table 4.0: Fastener Torque Requirements

Fastener Location / Application	Thread Size & Grade	Target Torque Value
Main Electrical Busbar Connections	M8, Grade 8.8 Brass/Cu	$14.5 \text{ Nm} \pm 0.5 \text{ Nm}$
Recirculation Fan Impeller Hub Locking Bolt	M16, Grade 10.9 Steel	$125.0 \text{ Nm} \pm 5.0 \text{ Nm}$
Conveyor Drive Sprocket Set Screws	M10, Grade 12.9 Steel	$38.0 \text{ Nm} \pm 1.5 \text{ Nm}$
Combustion Chamber Flange Bolts	M12, Grade 8.8 Steel	$65.0 \text{ Nm} \pm 2.0 \text{ Nm}$
Oven Casing Access Panel Fasteners	M6, Stainless A2-70	$8.5 \text{ Nm} \pm 0.5 \text{ Nm}$

6. SECTION 6: SAFETY-CRITICAL CORRECTIVE ACTIONS AND LOTO MANDATE

6.1 The following corrective actions involve hazardous energy sources (electrical, thermal, pneumatic, or mechanical) and are classified as Safety-Critical.

6.2 Execution of any safety-critical corrective action requires:

1. Full execution of the Vindhya Precision Works Lockout/Tagout (LOTO) procedure per corporate standard SOP-SAF-001.
2. Issuance of a verified Hot Work or Confined Space Entry Permit by the shift safety officer where applicable.
3. Verification of zero energy state using a calibrated multi-meter (CAT IV rated) before physical intervention.

6.3 Explicit List of Safety-Critical Corrective Actions:

- * SC-01: Replacement, wiring inspection, or internal adjustment of any Solid-State Relay (SSR) or main heating element contactor. (*Hazard: High Voltage / Electrocution*).
- * SC-02: Servicing, alignment, or tensioning of the internal conveyor drive chain and mechanical mesh assembly while access doors are open. (*Hazard: Mechanical Entanglement / Crush*).
- * SC-03: Maintenance, cleaning, or replacement of the natural gas burner manifold, pilot assembly, and automatic fuel gas shut-off valves. (*Hazard: Gas Leak / Explosion / Asphyxiation*).
- * SC-04: Entry into the oven chamber for internal duct cleaning or impeller maintenance prior to the internal ambient temperature dropping below 35.0°C and issuance of a confined space entry permit. (*Hazard: Severe Thermal Burns / Heat Stress*).
- * SC-05: Replacement of the recirculation fan motor (VPW-MOT-55) or servicing of its electrical supply leads. (*Hazard: Arc Flash / High Voltage*).

END OF PROCEDURE